

CHAPTER 15 SUMMARY

Electric Charges

(Section 15.1) Like charges repel; unlike charges attract.

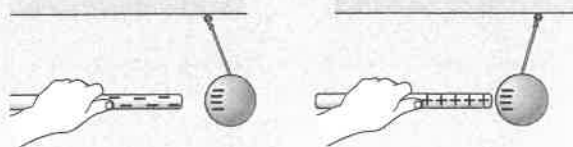
Charge is **conserved**: The sum of all positive and negative charge within a closed system is a constant.

Charge is **quantized**: Free charges always appear in integer multiples of the elementary charge e .

Charges of subatomic particles:

proton: $+e = +1.602 \times 10^{-19} \text{ C}$ neutron: charge = 0

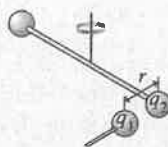
electron: $-e = -1.602 \times 10^{-19} \text{ C}$



Coulomb's Law

(Section 15.2) Coulomb's law governs the electrostatic force between point charges.

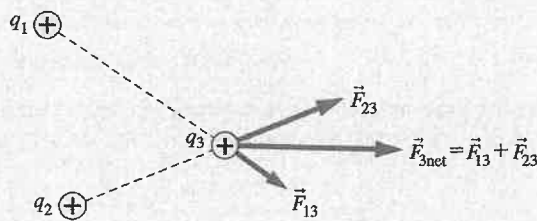
Coulomb's law: $F = \frac{k|q_1||q_2|}{r^2}$



Coulomb's Law for Multiple Charges

(Section 15.3) By the principle of **superposition**, the net electric force on a charge is the sum of the forces due to all the other charges.

Principle of superposition: $\vec{F}_{3\text{net}} = \vec{F}_{13} + \vec{F}_{23}$ for the three charges shown.



Electric Fields

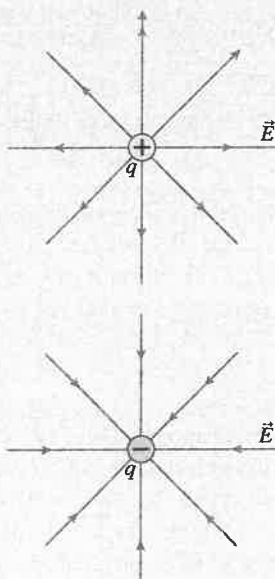
(Section 15.4) The **electric field** is a vector field that represents the force per unit charge acting on a test charge.

Electric field lines originate from positive charges and terminate on negative charges.

Electric field defined: $\vec{E} = \frac{\vec{F}}{q_0}$

Electric field of a point charge: $E = \frac{F}{q_0} = k \frac{|q|}{r^2}$

Electric field between a pair of oppositely charged planes: $E = \frac{\sigma}{\epsilon_0}$

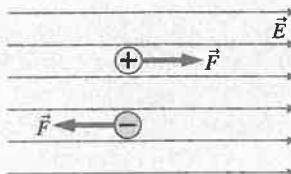


Charged Particles in Electric Fields

(Section 15.5) Charged particles are accelerated in electric fields.

The force on a positively charged particle is in the same direction as the electric field; the force on a negatively charged particle is in the opposite direction of the electric field.

Force on a charged particle in an electric field: $\vec{F} = q\vec{E}$



NOTE: Problem difficulty is labeled as ■ straightforward to ■■■ challenging. Problems labeled BIO are of biological or medical interest.

Conceptual Questions

- Suppose you bring a plastic rod near a plastic ball that is suspended by a thread. If the ball is repelled by the rod, what can you conclude about the net charge on the ball? Answer the same question if the ball is attracted to the rod.
- Pull off a long strip of transparent tape and hold it vertically with one hand. Bring your other hand near the hanging tape, and notice that the tape is attracted to your (electrically neutral) hand. Explain.
- Why is charge usually transferred by electrons rather than protons?
- List the similarities and differences between Coulomb's law and Newton's law of gravitation.
- An electron and proton are released from rest some distance apart. Will they meet halfway between their original positions? Explain.
- A small ball with charge Q is held fixed. A second identical ball (also with charge Q) is released some distance directly above the fixed ball. (a) Argue that the second ball will oscillate along a vertical line. (b) Is the oscillation simple harmonic motion? Explain.
- How is action at a distance (the force between two distant particles) explained using the exchange of other particles?
- Justify the rule that the electric field strength is proportional to the local density of electric field lines. *Hint:* Consider the electric field around a point charge, and look at the electric field lines passing through two concentric spheres of different radii. Then compare the density of field lines along the surface of each sphere.
- Can two electric field lines cross each other? Explain.
- Sketch the electric field lines generated by a pair of $-Q$ point charges.
- Sketch the electric field diagram in the region around the point charges $+Q$ and $-3Q$.
- Describe how the electric field appears if you are far away from the point charges $+Q$ and $-Q/2$ shown in Figure 15.19b.
- Argue why an electric dipole placed in a uniform electric field should oscillate with simple harmonic motion.
- Why doesn't the electric field strength near an infinite, uniformly charged plane vary depending on the distance from the plane?
- Which one of the following is true regarding a proton and an electron in identical electric fields? (a) The force on the electron is larger. (b) The force on the proton is larger. (c) The acceleration of the electron is larger. (d) The acceleration of the proton is larger.
- Three identical charges of $10.0\ \mu\text{C}$ are placed at the vertices of an equilateral triangle, that is $0.25\ \text{m}$ on a side. The net force acting on each charge has a magnitude (a) $57.6\ \text{N}$; (b) $28.8\ \text{N}$; (c) $24.9\ \text{N}$; (d) $14.4\ \text{N}$.
- A point charge $-1.2\ \mu\text{C}$ is $0.50\ \text{m}$ away from a second point charge $+1.0\ \mu\text{C}$. The force on a third charge, $+1.4\ \mu\text{C}$, placed exactly halfway between the other two is (a) $0.44\ \text{N}$, directed toward the negative charge; (b) $0.11\ \text{N}$, directed toward the negative charge; (c) $1.76\ \text{N}$, directed toward the negative charge (d) $0.11\ \text{N}$, directed toward the positive charge.
- The electric field halfway between two point charges, $-5.0\ \text{nC}$ and $+5.0\ \text{nC}$, separated by a distance of $0.30\ \text{m}$ is (a) zero; (b) $1000\ \text{N/C}$, directed toward the negative charge; (c) $2000\ \text{N/C}$, directed toward the negative charge; (d) $4000\ \text{N/C}$, directed toward the negative charge.
- A proton produces an electric field of exactly $500\ \text{N/C}$ at a distance of (a) $2.9 \times 10^{-12}\ \text{m}$; (b) $1.7 \times 10^{-6}\ \text{m}$; (c) $1.3 \times 10^{-3}\ \text{m}$; (d) $1.0\ \text{m}$.
- The electric field and gravitational field (a) have the same units; (b) both depend on mass; (c) are both attractive and repulsive forces; (d) both have inverse-square distance dependence.
- An alpha particle (produced in many nuclear reactions) has a mass of $6.64 \times 10^{-27}\ \text{kg}$ and a charge of $+2e$. In a uniform electric field inside a linear accelerator, an alpha particle is observed to have an acceleration of $6.0 \times 10^{13}\ \text{m/s}^2$. The magnitude of the electric field is (a) $1.25 \times 10^6\ \text{N/C}$; (b) $2.50 \times 10^6\ \text{N/C}$; (c) $5.00 \times 10^6\ \text{N/C}$; (d) $2.50 \times 10^7\ \text{N/C}$.

Problems

Section 15.1 Electric Charges

- (a) How much charge is contained in $1\ \text{kg}$ of electrons? (b) How much charge is contained in $1\ \text{kg}$ of protons?
- The Faraday is a unit of charge used in electrolysis and is defined as the charge of one mole of protons. Find the conversion factor between coulombs and Faradays.
- Estimate the number of electrons present in a (neutral) 1.0-kg block of carbon.
- Suppose Earth contains roughly an equal number of protons and neutrons. Estimate (a) the number of protons in Earth and (b) the charge Earth would have if all its electrons were removed.
- Suppose that the charge on the electron and proton did not have the same magnitude but instead differed by one part in 10^{10} . What would be the net charge on a 1.0-kg sample of helium gas?
- A plastic rod is rubbed with a cloth, and in the process it acquires a charge of $-2.3 \times 10^{-8}\ \text{C}$. Did the mass of the rod increase or decrease, and by how much?

Section 15.2 Coulomb's Law

- Two small charged balls have a repulsive force of $0.12\ \text{N}$ when they are separated by a distance of $0.85\ \text{m}$. The balls are moved closer together, until the repulsive force is $0.60\ \text{N}$. How far apart are they now?

Multiple-Choice Problems

- What is the net charge of a collection of 2.5×10^{18} electrons? (a) $0.025\ \text{C}$; (b) $0.40\ \text{C}$; (c) $2.5\ \text{C}$; (d) $4.0\ \text{C}$.
- The net charge of one mole of protons is (a) $1.60 \times 10^{-19}\ \text{C}$; (b) $1.00\ \text{C}$; (c) $1.60 \times 10^4\ \text{C}$; (d) $9.65 \times 10^4\ \text{C}$.
- What is the magnitude of the force acting on a proton at a distance $0.15\ \text{m}$ from another proton? (a) $1.1 \times 10^{-36}\ \text{N}$; (b) $1.0 \times 10^{-26}\ \text{N}$; (c) $6.4 \times 10^{-8}\ \text{N}$; (d) $1.5 \times 10^{-27}\ \text{N}$.
- Two charges lie on the $+x$ -axis: $12\ \text{nC}$ at $x = 0.24\ \text{m}$ and $16\ \text{nC}$ at $x = 0.64\ \text{m}$. Where on the x -axis could a third charge be placed so that the net force on it is zero? (a) There is no such place; (b) $x = 0.47\ \text{m}$; (c) $x = 0.43\ \text{m}$; (d) $x = 0.29\ \text{m}$.
- Two charged particles each experience a force of $20\ \text{N}$ when they are a distance d apart. The force will be $40\ \text{N}$ when the separation distance is changed to (a) $d/4$; (b) $d/2$; (c) $d/\sqrt{2}$; (d) $2d$.

34. ■ Treat as point charges two quarks of charge separated by a distance of in a proton, and find the electrostatic force between them.
35. ■■ Suppose all the electrons could be removed from Earth. (a) Find the force on a proton just above Earth's surface under these conditions. (b) Compare your answer in part (a) with the weight of the proton.
36. ■ A free electron and a free proton are exactly 1.0 cm apart. Find the magnitude and direction of (a) the acceleration of the proton and (b) the acceleration of the electron.
37. ■■ Two Ping-Pong balls each have a mass of 1.4 g and carry a net charge of $0.450 \mu\text{C}$. One ball is held fixed. At what height should the second ball be placed directly above the fixed ball if it is to remain at rest there?
38. ■■ **BIO Forces in a DNA molecule.** In a chemical reaction, the two ends of a $2.45\text{-}\mu\text{m}$ -long DNA molecule acquire charges of $+e$ and $-e$, respectively. What is the force between the two ends of the molecule?
39. ■ How far should an electron be from a proton so that the electrostatic force on the electron is just equal to the electron's weight? Comment on the significance of your result.
40. ■■ (a) Use the astronomical data in Appendix E to compute the gravitational force between Sun and Earth. (b) Suppose gravity were turned off, and you wanted to replace the gravitational force with an electrostatic force of the same magnitude you computed in part (a). If the Sun had charge $+Q$ and Earth $-Q$, determine the value of Q .
41. ■■ Two identical pendulums of length L hang side by side from a common support point, with their bobs (each having mass m) just touching. An equal charge Q is then placed on each bob. Find an equation relating θ (the angle each pendulum string makes with the vertical, as shown in Figure P15.41) to the other parameters given in the problem.

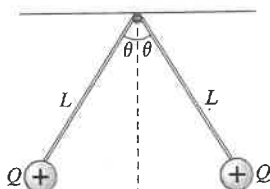


FIGURE P15.41

Section 15.3 Coulomb's Law for Multiple Charges

42. ■■ Three point charges are placed on the x -axis as follows: $20 \mu\text{C}$ at $x = 0$; $30 \mu\text{C}$ at $x = 0.50 \text{ m}$; and $-10 \mu\text{C}$ at $x = 1.0 \text{ m}$. Find the net force on each point charge.
43. ■■ Two point charges, $-25 \mu\text{C}$ and $-75 \mu\text{C}$, are held fixed exactly 1.0 m apart. (a) Where should a third charge be placed so that the net force on it is zero? (b) Explain why the magnitude and sign of the third charge do not affect your answer to part (a).
44. ■■ Two point charges, -10 nC and $+15 \text{ nC}$, are held fixed a distance 0.50 m apart. (a) Where can a $+20\text{-nC}$ charge be placed so that the net force on it is zero? (b) Repeat part (a) if the third charge is -20 nC .
45. ■■ Three identical charges of $2.0 \mu\text{C}$ are placed at the vertices of an equilateral triangle, 0.15 m on a side. Find the net force acting on each charge.
46. ■■ Three charges are fixed in the x - y plane as follows: 1.5 nC at the origin $(0, 0)$; 2.4 nC at $(0.75 \text{ m}, 0)$; -1.9 nC at $(0, 1.25 \text{ m})$. Find the force acting on the charge at the origin.
47. ■■ Four identical point charges of 5.6 nC are placed at the corners of a square, 0.25 m on a side. Find the force acting on each charge.
48. ■■ Four point charges are placed on the corners of a square of side d as shown in Figure P15.48. (a) Find the net force (magnitude and direction) on each charge. (b) If a fifth charge of size $+Q$ is brought into the picture and placed at the center of the square, what is the net force acting on it?
49. ■■ A point charge $+6.5 \mu\text{C}$ is placed at the origin, and a second charge $-4.2 \mu\text{C}$ is placed in the x - y plane at $(0.35 \text{ m}, 0.45 \text{ m})$. Where should a third charge be placed in the plane so that the net force acting on it is zero?
50. ■■ One electron is at the origin of a coordinate system. A second electron, somewhere in the x - y plane, feels a force $2.4 \times 10^{-22} \text{ N } \hat{i} + 3.6 \times 10^{-22} \text{ N } \hat{j}$. Where is the second electron located?
51. ■■ Three charges are fixed in the x - y plane as follows: 1.5 nC at the origin $(0, 0)$; 2.4 nC at $(0.75 \text{ m}, 0)$; -1.9 nC at $(0, 1.25 \text{ m})$. Find the force acting on the negative charge.

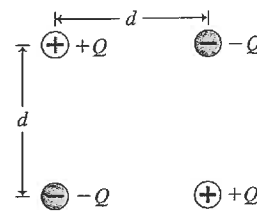


FIGURE P15.48

Section 15.4 Electric Fields

52. ■ Compute the electric field at the surface of (a) a proton, assuming the proton is a sphere of radius $1.2 \times 10^{-15} \text{ m}$; (b) a lead nucleus, which has charge $82e$, assuming the nucleus is a sphere of radius $7.1 \times 10^{-15} \text{ m}$.
53. ■ What is the electric field a distance 1.0 mm from an electron?
54. ■■ Two positive point charges, each $15 \mu\text{C}$, lie along the x -axis at $x = -0.15 \text{ m}$ and $x = +0.15 \text{ m}$. Find the electric field at (a) the origin $(0, 0)$ and (b) the point $(0, 0.20 \text{ m})$ on the y -axis.
55. ■■ For the point charges described in the preceding problem, rank in increasing order the magnitude of the electric field at the following positions: (a) $(0, 0)$; (b) $(0.25 \text{ m}, 0)$; (c) $(0.10 \text{ m}, 0)$; (d) $(0, 0.10 \text{ m})$.
56. ■■ Two point charges, $-25 \mu\text{C}$ and $+50 \mu\text{C}$, are separated by a distance of 20.0 cm. What is the electric field (magnitude and direction) at a point halfway between the two charges?
57. ■■ For the electric dipole in Example 15.7, compute the electric field at the following locations: (a) $(-15 \text{ cm}, 0)$; (b) $(0, -5 \text{ cm})$.
58. ■■ An electric dipole on the x -axis consists of a charge $-Q$ at $x = -d$ and a charge $+Q$ at $x = +d$. (a) Find a general expression (in terms of Q and d) for the electric field at points on the x -axis with $x > d$. (b) Show that for $x \gg d$ the magnitude of the field you computed in part (a) is *approximately* equal to $E = 4kQd/x^3$ and therefore approximates an *inverse-cube* distance law.
59. ■■ A pendulum consists of a 0.155-kg ball hanging vertically from a light thread. The ball has a net charge of $235 \mu\text{C}$. If the pendulum is placed into a uniform, horizontal electric field of 1250 N/C (Figure P15.59), find the angle the pendulum string makes with the vertical when the system is in equilibrium.

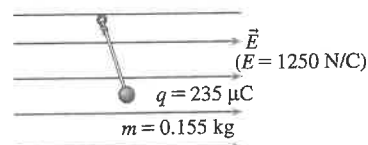


FIGURE P15.59

60. ■ A large plane having uniform charge density has an electric field just outside the plane that points directly toward the plane with magnitude $1.25 \times 10^4 \text{ N/C}$. What is the surface charge density?
61. ■ If a charged plane has a uniform surface charge density of $-1.5 \times 10^{-7} \text{ C/m}^2$, how many electrons will be on the surface in a square that is 1.0 mm on a side?
62. ■ For the charged plane described in the preceding problem, what is the electric field just outside the plane's surface?
63. ■ ■ A pair of charged, parallel planes are used to create a uniform electric field between the planes. The charge densities on the two planes are equal in magnitude but have opposite sign. Find the charge density on each plane if the electric field between them has a magnitude of $10,000 \text{ N/C}$.
64. ■ ■ A $23.6\text{-}\mu\text{C}$ point charge lies at the origin. Find the electric field at the following points in the x - y plane: (a) $(0.95 \text{ m}, 0)$; (b) $(1.25 \text{ m}, 0.75 \text{ m})$.
65. ■ Assume that the hydrogen atom consists of an electron in a circular orbit around a proton, with an orbital radius of $5.29 \times 10^{-11} \text{ m}$. (a) What is the electric field acting on the electron? (b) Use your answer in part (a) to find the force acting on the electron.
66. ■ ■ Four point charges lie in the x - y plane: $+Q$ at (a, a) ; $+2Q$ at $(-a, -a)$; $-Q$ at $(-a, a)$; and $-2Q$ at $(a, -a)$. Find the electric field at the origin.
67. ■ ■ Two point charges lie on the x -axis: $0.25 \mu\text{C}$ at $x = 0.25 \text{ m}$ and $0.16 \mu\text{C}$ at $x = 0.75 \text{ m}$. Find the place(s) where the electric field is zero.
68. ■ ■ ■ Two point charges lie on the x -axis: $1.65 \mu\text{C}$ at $x = 1.00 \text{ m}$ and $-2.30 \mu\text{C}$ at $x = 2.50 \text{ m}$. Find the place(s) where the electric field is zero.
69. ■ ■ In a particular region of space, there exists a uniform electric field of magnitude 12.4 kN/C in the $+x$ -direction. Now a $-1.0\text{-}\mu\text{C}$ charge is placed in this field, at a position we will call the origin $(0, 0)$. (a) Find the net electric field at the point $(0.50 \text{ m}, 0)$. (b) Find the net electric field at the point $(0, 0.50 \text{ m})$.
- negative plate, find (a) its speed when it reaches the positive plate and (b) the time it takes to travel across the 2.3-mm gap.
77. ■ ■ ■ An electron in an oscilloscope tube is traveling horizontally with a speed of $3.9 \times 10^6 \text{ m/s}$. It passes through a 2.0-cm -long region with an electric field pointing upward with a magnitude of 1500 N/C . What is the electron's velocity when it emerges from the electric field region?
78. ■ ■ In the preceding problem, how far downward (from its original trajectory) is the electron deflected during the time it's in the electric field?
79. ■ ■ In an x-ray tube, electrons are accelerated in a uniform electric field and then strike a metal target. Suppose an electron starting from rest is accelerated in a uniform electric field directed horizontally and having a magnitude of 2300 N/C . The electric field covers a region of space 10.5 cm wide. (a) What is the speed of the electron when it strikes the target? (b) How far does it fall under the influence of gravity during its flight?
80. ■ In an optics experiment, a discharge tube is filled with singly ionized argon atoms, each having a mass of $6.64 \times 10^{-26} \text{ kg}$. If the electric field inside the tube is 7500 N/C , find the acceleration of each argon ion.
81. ■ ■ An electron is traveling with a speed of $2.1 \times 10^6 \text{ m/s}$ when it enters a uniform electric field that is parallel to its direction of travel. After moving for another 9.2 cm , the electron comes to rest. (a) What is the magnitude of the electric field? (b) What is the direction of the electric field, relative to the electron's original motion?
82. ■ ■ A proton, initially traveling in the $+x$ -direction with a speed of $5.45 \times 10^5 \text{ m/s}$, enters a uniform electric field directed vertically upward. After traveling in this field for $3.92 \times 10^{-7} \text{ s}$, the proton's velocity is directed 45° above the $+x$ -axis. What is the strength of the electric field?

Section 15.5 Charged Particles in Electric Fields

70. ■ What is the acceleration of an electron that is placed in a uniform electric field of 6500 N/C pointing in the $+y$ -direction?
71. ■ A small bit of photocopier toner having mass of $3.5 \times 10^{-15} \text{ kg}$ carries a net charge of $2e$ and has an acceleration of 0.50 m/s^2 . What is the magnitude of the electric field causing the acceleration?
72. ■ Two parallel plates carry uniform charge densities -0.50 nC/m^2 and $+0.50 \text{ nC/m}^2$. (a) Find the electric field between the plates. (b) Find the acceleration of an electron between these plates.
73. ■ An electron moving in the x - y plane has an acceleration of $3.25 \times 10^{13} \text{ m/s}^2$ directed at an angle 30° above the $+x$ -axis. Find the electric field responsible for that acceleration.
74. ■ ■ Two large parallel plates are separated by 0.75 cm and carry uniform charge densities equal in magnitude and opposite in sign. A proton is between the plates and has an acceleration of magnitude $1.39 \times 10^{11} \text{ m/s}^2$. Find the charge density on the plates.
75. ■ ■ In a Millikan oil-drop experiment, using oil with a density of 940 kg/m^3 , a droplet with a charge of $-3e$ is just balanced against gravity in an electric field of 11.5 kN/C . (a) What is the direction of the electric field? (b) What is the size of the (spherical) droplet?
76. ■ ■ A pair of parallel plates are charged with uniform charge densities of -35 pC/m^2 and $+35 \text{ pC/m}^2$. The distance between the plates is 2.3 mm . If a free electron is released at rest from the
- General Problems**
83. ■ ■ A 2.1-g Ping-Pong ball is rubbed with a piece of wool, resulting in a net positive charge of $7.4 \mu\text{C}$ on the ball. How many electrons were transferred from the ball to the wool in this process?
84. ■ ■ You have three charges, two of which have a charge of $+4 \mu\text{C}$ and the other a charge of $-1 \mu\text{C}$. Show how you can place the three charges along a line so that there is no net force on any of them.
85. ■ ■ A charge of $3 \mu\text{C}$ is at the origin and a charge of $-2 \mu\text{C}$ is on the y -axis at $y = 50 \text{ cm}$. (a) Where can you place a third charge so that the force acting on it is zero? (b) What is the electric field at the location you found in part (a)?
86. ■ ■ Two charges, one twice as large as the other, are located 15 cm apart, and each experiences a repulsive force of 95 N from the other. (a) What is the magnitude of each charge? (b) Does your answer to part (a) depend on whether the charges are positive or negative? Why or why not? (c) At what location is the electric field zero?
87. ■ ■ In his famous experiment that demonstrated quantization of electric charge, Robert A. Millikan suspended small oil drops in an electric field. (a) With a field strength of 20 MN/C , what is the mass of a drop containing 10 elementary charges that can just be suspended in that field? (b) Indicate the direction of the electric field if the elementary charges are (i) positive; (ii) negative.
88. ■ ■ ■ A proton moving at $3.8 \times 10^5 \text{ m/s}$ to the right enters a region where a uniform electric field of magnitude 56 kN/C points

to the left. (a) How far will the proton travel before it stops moving? (b) What will happen to the proton after it stops moving? (c) Graph the proton's position versus time, velocity versus time, and acceleration versus time after it enters the field.

89. **B10 ■■ Electric bees.** As they fly through the air, honeybees can pick up a small amount of electric charge. Suppose a 0.12-g bee acquires a charge of -55 pC . (a) How many excess electrons has the bee acquired? (b) Find the electric force on the bee in the electric field just above Earth's surface, 120 N/C directed vertically downward. (c) Compare your answer in part (b) to the bee's weight.
90. **■■■** An electric dipole lies on the y -axis and consists of an electron at $y = 0.60 \text{ nm}$ and a proton at $y = -0.60 \text{ nm}$. Find the electric field (a) at a point halfway between the two charges; (b) at the point $x = 2.0 \text{ nm}, y = 0$; and (c) at the point $x = -2.0 \text{ nm}, y = 0$.
91. **■■■** Two identical small metal spheres carry charges q_1 and q_2 . When they're 1.0 m apart, they experience a 2.5-N attractive force. Then they are brought together so that charge moves from one sphere to the other until the net charges are equal. They are again placed 1.0 m apart and now they repel with a 2.5-N force. What are the original charges q_1 and q_2 ?

92. **■■■** Two 2.3-g Ping-Pong balls hang suspended, with the balls just touching when they hang from a common support. Each ball's center is initially 60 cm below the support. The balls are then each given an identical amount of charge q , and each string makes an angle of 15° with the vertical as shown in Figure GP15.92. Find the magnitude of the charge q .

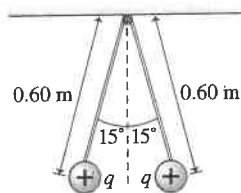


FIGURE GP15.92

93. **■■** (a) For the scanning electron microscope described in Example 15.11, find the electron's vertical displacement. (b) What would be the electron's vertical displacement if the scanning electric field were reduced to $3.21 \times 10^3 \text{ N/C}$ (half its original value)?
94. **■■■** Suppose you have a friend who has a part-time job translating computer maintenance training manuals from Japanese to English. A section of the manual for an inkjet printer is missing, and you need to replace the missing information. The printer

works by steering ink drops with charge q and mass m between two parallel surfaces where there is a uniform electric field of magnitude E that deflects them an appropriate amount. The field extends over a length L and width d , and each ink drop enters halfway between the charged plates as shown in Figure GP15.94. Find the minimum speed v the drop must have in order to make it through the region without striking one of the plates. *Note:* You may assume that the electric field is large enough that gravity can be neglected.

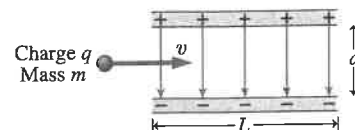


FIGURE GP15.94

95. **B10 ■■ DNA compression.** Due to its double-helix structure, a DNA molecule acts like a spring that can be compressed. Biophysicists have estimated the force constant for one type of DNA molecule to be 0.60 nN/nm . Find the compression of a $75\text{-}\mu\text{m}$ -long DNA molecule that acquires a charge of $+1.2 \text{ pC}$ at one end and -1.2 pC at the other.

Answers to Chapter Questions

Answer to Chapter-Opening Question

The conducting metal skin of the aircraft shields the interior. Free electric charges within the metal rearrange themselves quickly in the presence of external charge, in order to prevent penetration of the electric field.

Answers to GOT IT? Questions

Section 15.2 (e) The forces are repulsive, and the magnitudes of the two forces are the same.

Section 15.3 (c) Directed toward the right.

Section 15.4 (b) Directed toward the left.

Section 15.5 (d) In the opposite direction but smaller.